Markov Chains (AI HL)

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.03

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Are $\mathbf{u} = (2; 0; 0)$ and $\mathbf{w} = (7; 0; 0)$ parallel? Show your complete algebraic work.

Final Answer: Yes. Working through this vectors step: Check if $\mathbf{w} = k\mathbf{u}$ for a scalar k: $k = 3.5$ works, so they are parallel.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the number of ways to arrange the 7 letters in the word SUCCESS.

The word SUCCESS has 7 letters with 3 S's, 2 C's, 1 U, and 1 E. The number of distinct permutations is $\frac{7!}{3!\,2!} = \frac{5040}{6 \times 2} = \frac{5040}{12} = 420$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Show that $x^3 + 2x - 6 = 0$ has a root in $[1, 2]$. Taking $x_0 = 1.5$, perform one Newton-Raphson iteration.

$f(1) = -3 < 0$, $f(2) = 6 > 0$: sign change confirms root. $f(1.5) = 1.5^3 + 3 - 6 = -0.625$.
$f'(x) = 3x^2 + 2 \Rightarrow f'(1.5) = 8.75$. $x_1 = 1.5 - \frac{-0.625}{8.75} \approx 1.571$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A fair coin is tossed 10 times. Find the probability of getting (a) exactly 6 heads, (b) fewer than 3 heads.

(a) $\binom{10}{6}(0.5)^{10} = 210 \times \frac{1}{1024} \approx 0.2051$. (b) $P(X \le 2) = P(0) + P(1) + P(2) = \frac{1 + 10 + 45}{1024} = \frac{56}{1024} \approx 0.0547$.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Cars arrive at a toll booth at an average rate of 3 per minute. Find the probability that (a) exactly 5 arrive in one minute, (b) at least 2 arrive in 30 seconds.

(a) $P(X = 5) = \frac{e^{-3} \cdot 243}{120} \approx 0.1008$. (b) $\lambda = 1.5$: $P(X \ge 2) = 1 - P(0) - P(1) = 1 - e^{-1.5} - 1.5e^{-1.5} = 1 - 2.5e^{-1.5} \approx 0.4422$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Parametric & Polar · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Vector Calculus · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[About A2 Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

[Complex Numbers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Continuous Random Variables · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

[Differentiation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[Differential Equations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Exponentials & Logarithms · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Hypothesis Testing · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

[Integration · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Matrices (AI HL)

[Numerical Methods · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

[Poisson Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Proof by Mathematical Induction

[Sampling & Estimation · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

[Advanced Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

[About AS Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a standard transition probability matrix T (where entry T_{ij} represents the probability of moving from state i to state j), what must the sum of the entries in each row equal?

a. 1 ✓

b. 0

c. The number of states

d. $\frac{1}{n}$

Because each row corresponds to the probability distribution of transitioning from a given state to all possible next states, the sum of probabilities across each row is strictly 1.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given initial state probability vector $\mathbf{s}_0 = [0.6, 0.4]$ and transition matrix $T = [[0.8, 0.2], [0.3, 0.7]]$, what is the state distribution after one step $\mathbf{s}_1 = \mathbf{s}_0 T$?

a. [0.60, 0.40] ✓

b. [0.55, 0.45]

c. [0.66, 0.34]

d. [0.48, 0.28]

$\mathbf{s}_1 = [0.6(0.8) + 0.4(0.3), 0.6(0.2) + 0.4(0.7)] = [0.48 + 0.12, 0.12 + 0.28] = [0.60, 0.40]$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the Markov property (memoryless property) in stochastic processes?

a. The future state depends only upon the current state, not on the sequence of states that preceded it ✓

b. The process has zero variance

c. All states have equal probability of occurrence

d. The chain can never visit the same state twice

A first-order Markov chain satisfies $P(X_{n+1} = s \mid X_n, X_{n-1}, \ldots, X_0) = P(X_{n+1} = s \mid X_n)$, meaning the conditional probability distribution depends only on the present state.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How is the probability distribution $\mathbf{s}_n$ after n transitions computed from the initial state vector $\mathbf{s}_0$ and transition matrix T?

a. $\mathbf{s}_n = \mathbf{s}_0 T^n$ ✓

b. $\mathbf{s}_n = \mathbf{s}_0 + nT$

c. $\mathbf{s}_n = T^n \mathbf{s}_0$

d. $\mathbf{s}_n = (\mathbf{s}_0 T)^n$

Iterating the transition step $\mathbf{s}_1 = \mathbf{s}_0 T$, $\mathbf{s}_2 = \mathbf{s}_1 T = \mathbf{s}_0 T^2$ leads to $\mathbf{s}_n = \mathbf{s}_0 T^n$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What does entry $(T^3)_{2,1}$ represent?

a. The probability of transitioning from State 2 to State 1 in exactly 3 steps ✓

b. The probability of transitioning from State 1 to State 2 in 3 steps

c. Three times the one-step probability

d. The cube of the probability from State 2 to State 1

In the n-step transition matrix T^n, the entry in row i, column j is the conditional probability of being in state j after n steps given starting in state i.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What defining matrix equation must the steady-state probability vector $\boldsymbol{\pi} = [\pi_1, \pi_2]$ satisfy?

a. $\boldsymbol{\pi} T = \boldsymbol{\pi}$ **and** $\sum \pi_i = 1$ ✓

b. $T\boldsymbol{\pi} = \mathbf{0}$

c. $\boldsymbol{\pi} T = \mathbf{0}$

d. $\boldsymbol{\pi} = T^{-1}$

A steady-state distribution is stationary under transitions ($\boldsymbol{\pi} T = \boldsymbol{\pi}$) and its components form a valid probability distribution (sum to 1).

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the stationary distribution for $T = [[0.7, 0.3], [0.2, 0.8]]$.

a. $\boldsymbol{\pi} = [0.4, 0.6]$ ✓

b. $\boldsymbol{\pi} = [0.5, 0.5]$

c. $\boldsymbol{\pi} = [0.3, 0.7]$

d. $\boldsymbol{\pi} = [0.6, 0.4]$

Set $[\pi_1, \pi_2][[0.7, 0.3], [0.2, 0.8]] = [\pi_1, \pi_2] \Rightarrow 0.7\pi_1 + 0.2\pi_2 = \pi_1 \Rightarrow 0.2\pi_2 = 0.3\pi_1 \Rightarrow 3\pi_1 = 2\pi_2$. Using $\pi_1 + \pi_2 = 1$:
$\pi_1 + 1.5\pi_1 = 1 \Rightarrow 2.5\pi_1 = 1 \Rightarrow \pi_1 = 0.4$, $\pi_2 = 0.6$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

When is a Markov chain guaranteed to have a unique stationary distribution that is independent of the initial state?

a. **When the transition matrix is regular (some power T^k has strictly positive entries)** ✓

b. Only if the matrix is symmetric

c. Whenever $\det(T) = 0$

d. Only if there are 2 states

By the Perron-Frobenius theorem, a regular Markov chain has a unique stationary distribution, and $\lim_{n \to \infty} \mathbf{s}_0 T^n = \boldsymbol{\pi}$ for any initial distribution $\mathbf{s}_0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is an absorbing state in a Markov chain?

a. A state that, once entered, cannot be left ($T_{ii} = 1$ and $T_{ij} = 0$ for $j \neq i$) ✓

b. A state that has probability zero of being visited

c. A state that transitions with equal probability to all other states

d. A state where the steady-state probability is 0

An absorbing state has a loop transition probability of 1 to itself, trapping the process indefinitely once entered.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In an absorbing Markov chain in canonical form $P = [[I, 0], [R, Q]]$, what does the fundamental matrix $N = (I - Q)^{-1}$ calculate?

a. The expected number of visits to each transient state before absorption ✓

b. The probability of absorption in one step

c. The eigenvalues of the transition matrix

d. The total number of absorbing states

The fundamental matrix $N = (I - Q)^{-1}$ provides the expected time (number of steps) spent in each transient state starting from any transient state before reaching an absorbing state.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How is the absorption probability matrix B calculated from fundamental matrix N and transient-to-absorbing matrix R?

a. $B = NR$ ✓

b. $B = RN$

c. $B = N + R$

d. $B = (I - Q)R$

The matrix of absorption probabilities from each transient state into each absorbing state is given by the matrix product $B = NR$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
Triple integral $\iiint_V 1\,dV$ gives:

a. **volume of V** ✓

b. 0

c. surface area

d. 1

Applying Multiple Integrals principles, Triple integral of 1 = volume. Therefore, the verified correct option is volume of V.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)
Use polar coordinates to evaluate $\iint_R \sqrt{x^2+y^2}\,dA$, where R is the annulus $1 \le x^2+y^2 \le 4$.

a. $\frac{14}{3}$

b. $\frac{14\pi}{3}$ ✓

c. $\frac{28\pi}{3}$

d. $\frac{7\pi}{3}$

With $x = r\cos\theta,\ y = r\sin\theta$ and $dA = r\,dr\,d\theta$, the integral becomes elementary; its value is $\frac{14\pi}{3}$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
Optimize area of rectangle with perimeter 20: max area $=$

a. 25 ✓

b. 20

c. 50

d. 100

Applying Application of Derivatives principles, 5×5 square: area $= 25$. Therefore, the verified correct option is 25.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)
Riemann sum step width $\Delta x =$

a. $1/n$

b. $(b-a)/n$ ✓

c. $n(b-a)$

d. $(f(b)-f(a))/n$

Applying Fundamental Theorem principles, $(b-a)/n$ for n subintervals. Therefore, the verified correct option is $(b-a)/n$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Volume of cone $z = \sqrt{x^2 + y^2}$, $z \leq 1$ in cylindrical:

a. $2\pi/3$

b. $\pi/3$ ✓

c. π

d. $\pi/2$

Applying Multiple Integrals principles, $V = 2\pi\int_0^1 (1 - r)r\,dr = \pi/3$. Therefore, the verified correct option is $\pi/3$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the critical point of $f(x) = x^2 - 4x + 3$.

a. 1

b. 3

c. **2** ✓

d. -2

Applying Application of Derivatives principles, $f'(x) = 2x - 4 = 0 \Rightarrow x = 2$. Therefore, the verified correct option is 2.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Proof by Mathematical Induction

By the Intermediate Value Theorem, does $f(x) = x^3 - x - 1$ have a root in $[1, 2]$?

a. No — $f(1)$ and $f(2)$ have the same sign

b. No — f is not continuous on $[1, 2]$

c. Yes — $f(1)$ and $f(2)$ have opposite signs ✓

d. Cannot be determined without more information

$f(1) \approx -1.00$ and $f(2) \approx 5.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(1, 2)$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

The moment of inertia about the z-axis: $I_z = \iiint_V (x^2 + y^2)\rho\, dV$. For a uniform cylinder $r^2 \le R^2$, $0 \le z \le h$, $\rho = \text{const}$:

a. $\rho\pi R^2 h/2$

b. $\rho\pi R^2 h \cdot R^2/2$ ✓

c. $\pi R^2 h$

d. $\rho R^4 h$

$I_z = \rho \int_0^h \int_0^{2\pi} \int_0^R r^2 \cdot r\, dr\, d\theta\, dz = \rho h \cdot 2\pi \cdot R^4/4 = \rho\pi R^4 h/2 = mR^2/2$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Volume of spherical shell $1 \le \rho \le 2$:

a. 4π

b. $28\pi/3$ ✓

c. 8π

d. $4\pi/3$

Applying Multiple Integrals principles, $V = (4\pi/3)(8 - 1) = 28\pi/3$. Therefore, the verified correct option is $28\pi/3$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.04 Partial Fractions

For a region symmetric about the y-axis, $\iint_D x\, dA =$

a. 0 ✓

b. -1

c. 1

d. $A(D)$

Applying Multiple Integrals principles, By symmetry: $\iint x\, dA = 0$. Therefore, the verified correct option is 0.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

A rectangle has perimeter 40. Find the width w that maximises its area.

a. **10** ✓

b. 20

c. 5

d. $\frac{40}{3}$

With length $\ell = \frac{40}{2} - w$, $A(w) = w(\frac{40}{2} - w)$; $A'(w) = \frac{40}{2} - 2w = 0 \Rightarrow w = \frac{40}{4} = 10$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

$\int_0^1 \int_0^x y\,dy\,dx$: inner integral gives $x^2/2$, so outer integral =

a. 1

b. 1/4

c. 1/2

d. 1/6 ✓

Applying Multiple Integrals principles, $\int_0^1 x^2/2\,dx = 1/6$. Therefore, the verified correct option is $1/6$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

First step in proving $\sqrt{3}$ is irrational by contradiction:

a. assume $\sqrt{3}$ is rational and never simplify

b. assume $\sqrt{3} = p/q$ in lowest terms ✓

c. square both sides immediately

d. assume p, q share a common factor

A proof by contradiction begins by assuming the opposite (rational) in lowest terms, then derives a contradiction.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Moment of inertia about z-axis of uniform ball radius R, mass m:

a. $2mR^2/5$ ✓

b. $2mR^2/3$

c. $mR^2/2$

d. mR^2

Applying Multiple Integrals principles, Ball: $I_z = 2mR^2/5$. Therefore, the verified correct option is $2mR^2/5$.

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